

CNG 2057 Project: Part One

Executive Summary

Ryan Verrot

Pikes Peak State College

CNG 2057 1N1: Network Defense and Countermeasures

Professor [REDACTED]

2026-03-15 23:59

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This report provides a comprehensive threat intelligence analysis of three Advanced Persistent Threat (APT) groups currently posing a significant risk to U.S. defense and government sectors. The analysis identifies Volt Typhoon (China), APT29 (Russia), and APT33 (Iran) as high-priority adversaries based on their confirmed targeting of U.S. critical infrastructure, defense industrial base, and government networks between 2022 and 2025. Using the MITRE ATT&CK framework, this report categorizes adversary techniques by risk level and prioritizes mitigation strategies to address the most critical threats facing enterprise security operations.

Section A: Overview of Advanced Persistent Threats (APT)

The following three APT groups were selected based on their documented history of targeting U.S. government entities and defense industry organizations, their attribution to nation-state intelligence services, and their recent confirmed operational activity between 2022-2025:

- **Volt Typhoon (G1017) (China)**
 - ***Volt Typhoon Overview:***
 - Volt Typhoon, also tracked as Bronze Silhouette and Vanguard Panda, is attributed with high confidence to People's Republic of China (PRC) state-sponsored actors. This group targets U.S. critical infrastructure supporting military mobilization, including communications, manufacturing, utility, and transportation sectors in the continental United States and Guam. Volt Typhoon represents a strategic shift in Chinese cyber operations from intellectual property theft to operational preparation of the environment for potential infrastructure disruption. The group was selected due to its unique "Living off the Land" techniques that evade traditional detection methods, posing a high disruption risk to military mobilization capabilities.

- ***Volt Typhoon Capability:***
 - Volt Typhoon employs Living-Off-The-Land (LOTL) techniques, relying heavily on built-in Windows tools (PowerShell, WMI, netsh) and valid account credentials to avoid endpoint detection, while routing traffic through compromised SOHO routers to mask attribution (**CISA, 2023**).
- ***Volt Typhoon Motivation:***
 - The group's objective is strategic pre-positioning within critical infrastructure to enable disruption of communications between the U.S. and Asia during a potential military conflict (**Microsoft, 2023**).
- ***Volt Typhoon Recent Activity:***
 - In early 2024, the FBI and CISA disrupted the KV-botnet infrastructure used by Volt Typhoon, though the group successfully rebuilt its operational capability and continued targeting critical infrastructure throughout 2024-2025 (**DOJ Press Release, 2024**).
- **APT29 (G0016) (Russia)**
 - ***APT29 Overview:***
 - APT29, also known as Cozy Bear, Midnight Blizzard, and NOBELIUM, is attributed with high confidence to Russia's Foreign Intelligence Service (SVR). This group targets U.S. government agencies, diplomatic entities, and technology service providers through sophisticated supply chain compromises and cloud infrastructure exploitation. APT29 is best known for the 2020 SolarWinds supply chain attack, which compromised multiple U.S. federal civilian agencies including the Department of Homeland Security, State Department, Treasury, and Commerce. The group was selected due to its

persistent targeting of U.S. supply chains and cloud environments, representing a high espionage risk with advanced stealth capabilities.

- **APT29 Capability:**

- APT29 demonstrates advanced tradecraft in supply chain compromise (T1195), cloud identity exploitation using Golden SAML token forgery, and long-term persistence in enterprise networks (**CISA, 2021**).

- **APT29 Motivation:**

- The group's strategic objective is high-level intelligence gathering on U.S. foreign policy, sanctions discussions, and defense planning to inform Kremlin decision-making (**Microsoft Threat Intelligence, 2024**).

- **APT29 Recent Activity:**

- During 2023-2024, APT29 conducted the Midnight Blizzard campaign, breaching Microsoft's corporate email systems and targeting Hewlett Packard Enterprise, demonstrating continued focus on technology service providers and cloud infrastructure (**Microsoft Threat Intelligence, 2024**).

- **APT33 (G0064) (Iran)**

- **APT33 Overview:**

- APT33, also known as Elfin, Refined Kitten, Holmium, and Peach Sandstorm, is attributed with high confidence to Iran's Islamic Revolutionary Guard Corps (IRGC). This group targets U.S. aerospace and defense industries as well as the energy sector, conducting both espionage operations to steal proprietary technical data and maintaining the capability for destructive attacks. APT33 focuses on stealing technical specifications related to aviation technologies to support Iran's domestic drone and unmanned aerial vehicle programs. The

group was selected for its dual mission of intellectual property theft and destructive capability, representing a high risk to the defense industrial base.

- ***APT33 Capability:***

- APT33 employs extensive password spraying campaigns against cloud authentication portals (Outlook Web Access, VPN gateways) and sophisticated spear-phishing operations using fake job recruitment lures impersonating major defense contractors such as Boeing and Northrop Grumman (**Mandiant, 2017**).

- ***APT33 Motivation:***

- The group's strategic objectives include bridging Iran's technological gap in military aviation and drone capabilities while maintaining a cyber deterrent through the threat of destructive operations (**Microsoft Threat Intelligence, 2023**).

- ***APT33 Recent Activity:***

- Throughout 2023-2025, APT33 (tracked as Peach Sandstorm by Microsoft) shifted tactics toward identity-centric infiltrations, conducting large-scale password spraying against Microsoft 365 and Azure AD environments targeting U.S. satellite, defense, and pharmaceutical sectors (**Microsoft Threat Intelligence, 2023**).

Section B: Technique Risk Categorization

The following risk categorization aggregates the MITRE ATT&CK techniques observed across Volt Typhoon, APT29, and APT33. The techniques are classified and scored as High-Risk when used by all three APT groups, Moderate-Risk when used by two APT groups, and Low-Risk when used by only one of the APT groups. This methodology identifies the most commonly utilized adversarial techniques that represent the most critical defense gaps requiring immediate attention and remediation.

High-Risk Techniques (Utilized by All APT Groups Selected)

- ***T1078: Valid Accounts***
 - All three APT groups (Volt Typhoon, APT29, and APT33) prioritize stealing and using legitimate credentials to bypass detection systems and blend in with normal, everyday user activity.
- ***T1059: Command and Scripting Interpreter***
 - All three of the selected groups employ extensive use of PowerShell, CMD, and Unix shell commands to execute malicious code while avoiding custom malware deployments.
- ***T1003: OS Credential Dumping***
 - All groups practice credential dumping from LSASS memory or SAM databases to escalate privileges and enable lateral movement.
- ***T1027: Obfuscated Files or Information***
 - The trio of APT groups all make use of encoding (e.g., Base64), packing, and obfuscation techniques in an effort to hide malicious scripts and payloads from security tools.
- ***T1105: Ingress Tool Transfer***
 - After establishing initial access to victim networks, all three APT groups download additional tools, scripts, and/or malware from external servers.
- ***Detailed High-Risk Adversarial Techniques Information***
 - More details on the aforementioned High-Risk Adversarial Techniques can be found utilizing the MITRE ATT&CK Navigator tool with the verrot-ryan_consolidated-navigator-layers.json file attached to this executive summary with the MITRE ATT&CK Navigator tool. Selecting and right-clicking on a High-Risk Adversarial Technique from the Navigator layer provides a variety of options for additional information on the

technique. High-Risk techniques are denoted with a pastel red color (Color Hex Code #ff6666ff).

Moderate-Risk Techniques (Utilized by 2 of the 3 APT Groups Selected)

- ***T1566: Phishing***
 - Phishing is heavily utilized by APT29 and APT33 as a primary initial access vector. Volt Typhoon occasionally makes use of phishing, but primarily exploits public facing applications.
- ***T1090: Proxy***
 - Used by APT29 and Volt Typhoon to route command-and-control traffic through intermediary systems and hide the origin of attacks.
- ***T1110: Brute Force***
 - APT29 and APT33 employ password spraying campaigns to gain initial access to cloud authentication portals and VPN gateways.
- ***T1021: Remote Services***
 - Both APT29 and APT33 have been observed abusing RDP and SMB protocols for lateral movement within networks they have compromised.
- ***T1082: System Information Discovery***
 - Volt Typhoon and APT29 have used System Information Discovery to enumerate system configurations, installed software, and network topology during reconnaissance.
- ***Additional Moderate-Risk Adversarial Techniques from MITRE ATT&CK Navigator***
 - While conducting research for this executive summary, a modest list of 14 Moderate-Risk techniques associated with at least two of the chosen APT groups were identified. A comprehensive list of Moderate-Risk Adversarial Techniques can be found in the verrot-ryan_consolidated-navigator-layers.json file attached to this executive summary.

Moderate-Risk techniques are denoted using a medium-light yellow color (Color Hex Code #ffe766ff).

Low-Risk Techniques (Utilized by 1 of the 3 APT Groups Selected)

- ***T1485: Data Destruction***
 - Observed in APT33 operations; Iranian threat actors have demonstrated a willingness to deploy disk-wiping malware for destructive operations against targets.
- ***T1195: Supply Chain Compromise***
 - This is one of APT29's signature techniques, as observed during the SolarWinds attack, and is highly specific to SVR operational tradecraft.
- ***T1047: Windows Management Instrumentation (WMI)***
 - Commonly observed as a primary execution method in Volt Typhoon operations as part of their operational Living-Off-the-Land approach.
- ***T1204: User Execution***
 - Seen in APT33 spear-phishing campaigns that rely on users clicking through malicious links and/or opening weaponized attachments.
- ***T1588: Obtain Capabilities***
 - Observed in APT33's operations, the group has been documented acquiring malware and exploits rather than developing their own in-house tools.
- ***Additional Low-Risk Adversarial Techniques from MITRE ATT&CK Navigator***
 - While conducting research for this executive summary, an extensive list of over 50 low-risk techniques associated with at least one of the chosen APT groups were identified. A comprehensive list of Low-Risk Adversarial Techniques can be found in the verrot-ryan_consolidated-navigator-layers.json file attached to this executive summary. Low-Risk techniques are denoted using a vibrant green color (Color Hex Code #8ec843ff).

Section C: Techniques Posing the Highest Risk

The following five adversarial techniques from the High-Risk category are prioritized from the greatest to least risk posed to organizations based on the prevalence of their use across all three APT groups, their effectiveness at bypassing modern security controls, and current cyber threat intelligence reporting on enterprise security threats and trends:

1. T1078: Valid Accounts (Highest Priority)

Valid account abuse represents the most critical threat because all three APT groups prioritize utilizing this technique to bypass malware detection systems and blend in with legitimate traffic and user activity. When these adversaries authenticate using stolen credentials, endpoint protection tools and network monitoring systems often fail to detect malicious activity because the traffic appears to originate from authorized users on the network (**CrowdStrike, 2024**). Specifically, Volt Typhoon's operational model relies on valid accounts combined with Living-Off-The-Land techniques to maintain long-term persistence in critical infrastructure without deploying custom malware (**CISA, 2023**). Likewise, APT29 has demonstrated advanced capabilities in this domain through token theft and Golden SAML attacks that bypass multi-factor authentication (MFA) systems (**Microsoft Threat Intelligence, 2024**).

Both the 2023 MITRE Engenuity ATT&CK Evaluations and the 2024 Verizon Data Breach Investigations Report both identify credential abuse as the leading attack vector in successful enterprise compromises. This requires immediate mitigation and implementation of phishing-resistant MFA such as FIDO2 or hardware security keys for all access points (**CISA, 2023**). Additionally, the enforcement of strict least-privilege access policies to limit the impact of compromised credentials (**NIST, 2020**).

2. T1059: Command and Scripting Interpreter (2nd Priority)

The use of built-in scripting interpreters such as PowerShell, Windows Command Shell, and Unix shells represents the second-highest priority because it enables the three APT groups to execute

malicious code without deploying detectable malware binaries. This Living-Off-The-Land approach is a hallmark of modern APT operations, but particularly Volt Typhoon's operational tradecraft (**CISA, 2023**). All three of the APT groups leverage native system tools to avoid triggering any signature-based detection methods and to blend their activities with legitimate administrative tasks (**Mandiant, 2024**).

The 2024 CrowdStrike Global Threat Report identifies a 60% increase in attacks leveraging built-in system utilities rather than custom, in-house developed malware. Specifically, APT29 uses PowerShell extensively for credential theft and lateral movement through systems (**Microsoft Threat Intelligence, 2024**). On the other hand, APT33 utilizes scripting for reconnaissance and data exfiltration (**Microsoft Threat Intelligence, 2023**). Mitigation strategies include enabling PowerShell Constrained Language Mode to restrict the use of particularly dangerous cmdlets, implementing comprehensive Script Block Logging to capture all executed commands, and tuning Endpoint Detection and Response (EDR) tools to detect anomalous script execution patterns instead of relying entirely on malware signatures (**MITRE D3FEND, n.d.**).

3. T1003: OS Credential Dumping (3rd Priority)

OS credential dumping from operating system memory and databases represents our third-highest priority adversarial technique because it enables privilege escalation and lateral movement across enterprise networks. Volt Typhoon, APT29, and APT33 all employ this technique to extract credentials from LSASS memory, SAM databases, and cached domain credentials (**Mandiant, 2024**). Once adversaries have obtained administrative credentials, they can move laterally to high-value targets including domain controllers, file servers, and cloud management consoles. APT29 is especially sophisticated in this area as demonstrated by their utilization of custom tools to extract credentials while evading detection and maintaining persistence (**Microsoft Threat Intelligence, 2024**).

The SANS 2024 Threat Landscape Survey observed credential dumping in almost 78% of successful network compromises. Mitigation and remediation calls for the implementation of Credential

Guard on Windows systems to protect LSASS memory and deployment of Local Administrator Password Solution (LAPS) to prevent lateral movement using shared local administrator accounts (**Microsoft Security, 2023**), as well as monitoring for suspicious access to credential stores using telemetry from EDR solutions and SIEM correlation rules (**MITRE D3FEND, n.d.**).

4. T1027: Obfuscated Files or Information (4th Priority)

Obfuscation techniques are our fourth-highest priority adversarial technique because all three of the selected APT groups utilize encoding, packing, and encryption to hide malicious payloads from security scanning tools used in enterprise environments (**Nextron Systems, 2025**). APT29 specifically uses sophisticated obfuscation in its supply chain attacks to hide back doors within legitimate software updates (**Microsoft Threat Intelligence, 2024**). On the other hand, Volt Typhoon obfuscates PowerShell commands they utilize using Base64 encoding to evade command-line logging and detection (**Picus Security, 2024**). Finally, APT33 employs packing and encryption to protect their custom malware families from reverse engineering and signature-based detection (**Microsoft Threat Intelligence, 2023**).

The 2024 Mandiant M-Trends Report noted a 45% increase in the use of obfuscation techniques used by threat actors to evade endpoint protection platforms and solutions. Effective mitigation of obfuscation techniques requires the implementation of behavioral analysis capabilities that can detect malicious activity regardless of obfuscation, deployment of application control policies designed to prevent the execution of unsigned or unknown binaries, and the utilization of detonation sandboxes to analyze suspicious files in isolated environments before allowing them to operate on production systems (**Cybaverse, 2025**).

5. T1105: Ingress Tool Transfer (Lowest Priority)

The transfer of additional tools and malware after initial compromise ranks as our lowest-priority among the five High Risk techniques because it enables adversaries to expand their attack capabilities and adapt to the targeted environment. The three APT groups selected all download

secondary payloads, reconnaissance tools, and credential theft utilities after establishing initial access **(Mandiant, 2024)**. Specifically, APT29 utilizes this technique to deploy custom backdoors and

persistence mechanisms following system compromise **(Microsoft Threat Intelligence, 2024)**.

Additionally, Volt Typhoon transfer Living-Off-the-Land binaries and scripts to compromised systems to maintain operational capabilities **(Microsoft Threat Intelligence, 2023)**. Finally, APT33 downloads

additional malware families to support both espionage and destructive capabilities in their operations **(Microsoft Threat Intelligence, 2023)**.

The MITRE ATT&CK framework identifies ingress tool transfer as a critical technique empowering APTs because it allows adversaries to maintain a minimal footprint while retaining the ability to escalate attack operations **(MITRE ATT&CK, n.d.)**. Some mitigation strategies include implementing egress filtering and proxy controls to restrict outbound connections to infrastructure known to be malicious, deploying network-based intrusion prevention systems (IPS) to detect and block tool downloads, and monitoring for unusual file transfers using network traffic analysis and telemetry **(NIST, 2009)**.

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